

ANDOR VÁRI-KAKAS

EDUCATION

ETH Zürich , <i>Ph.D. in Computer Science</i>	Apr. 2026 – Mar. 2030 , Zürich, Switzerland
• Group: Advised by Prof. David Steurer • Areas of interest: High-dimensional problems, learning theory, average-case algorithms	
ETH Zürich , <i>M.Sc. in Computer Science, ESOP Excellence Scholar</i>	Sept. 2022 – May 2025 , Zürich, Switzerland
• Specialisation: Theoretical Computer Science major, Machine Learning minor • GPA: 5.91/6 • Relevant Modules: Advanced Algorithms, Statistical Learning Theory, Guarantees for ML, Probabilistic Methods in Combinatorics, Optimisation for DS, Network Modelling, Reliable and Trustworthy AI • Teaching Assistant: Computational Intelligence Lab, Advanced Graph Algorithms and Optimization, Advanced Algorithms	
University of Cambridge , <i>B.A. in Computer Science</i>	Oct. 2019 – Jun. 2022 , Churchill College, Cambridge, UK
• Grade: First class – rank 2/123 in 2nd year, 7/121 in 3rd year • Thesis: Reinforcement Learning Meets Balls into Bins, with Thomas Sauerwald • Relevant Modules: Randomised Algorithms, Bayesian Inference, Category Theory, Bioinformatics, Deep Neural Networks	

RESEARCH EXPERIENCE

Roche , <i>Machine Learning Researcher for Drug Discovery Intern</i>	Jun. 2025 – Dec. 2025 , Basel, Switzerland
• Developed algorithms for multi-objective LLM alignment and domain generalisation in antibody binding affinity prediction • Resulting paper MGDA-DECOUPLED: Geometry-Aware Multi-Objective Optimisation for DPO-based LLM Alignment accepted to ICLR 2026 AFAA Workshop	
ETH Zürich – Master’s Thesis , <i>Advisor: Prof. David Steurer</i>	Oct. 2024 – Apr. 2025 , Zürich, Switzerland
• Developed novel algorithm and corresponding meta-theorem characterising recoverability of k-colorings in one-sided expanders • Resulting paper Finding Colorings in One-Sided Expanders accepted to FOCS 2025	
ETH Zürich – Student Researcher , <i>Projects with various research groups</i>	Jan. 2023 – May 2024 , Zürich, Switzerland
• Fairness-accuracy tradeoffs in fairness-aware learning, with Fanny Yang • Lower and upper bounds on the size-Ramsey numbers of 1-hypertrees, with Angelika Steger • Statistical-computational gaps of mixed-sparse PCA, with Afonso Bandeira	
SZTAKI , <i>Machine Learning Researcher Intern</i>	Jun. 2021 – Sep. 2021 , Budapest, Hungary
• Worked on improving lung cancer classification on low-dose CT using the sinogram domain	
University of Oradea , <i>Student Research Assistant</i>	May 2021 – Jun. 2021 , Oradea, Romania
• Co-authored the Issues Related to the Use of Blockchains in IoT Applications paper presented at ICEMES 2021	
Semmelweis University , <i>Student Research Assistant</i>	Jul. 2018 – Jul. 2018 , Budapest, Hungary
• Performed statistical analysis on gene expression data to identify potential biomarkers for predicting tumour metastasis	

RECENT WORK EXPERIENCE

Optiver , <i>Quantitative Researcher Intern</i>	Jul. 2024 – Aug. 2024 , Amsterdam, Netherlands
• Learnt option theory, traded on real markets, did a project on auction inefficiencies	
QuantCo , <i>Data Science Working Student</i>	Mar. 2024 – Jun. 2024 , Zürich, Switzerland
• Optimised cross-fitting strategies for training causal inference metalearners	
Hudson River Trading , <i>Algo Developer Intern</i>	Jul. 2022 – Sep. 2022 , New York, US & London, UK
• Built algorithms for trading lead-lag products, analysed the reliability of rare-event strategies	
Blackrock , <i>Software Engineer Intern</i>	Jun. 2020 – Sep. 2020 , Budapest, Hungary
• Implemented a research framework for analysing portfolios	

SUBJECT-RELATED ACTIVITIES

Mediterranean Machine Learning Summer School	Transformers, RLHF, diffusion models, optimisation	2024
Hungarian Linguistics Olympiad	Co-organiser, website coordinator, team leader of Hungary at IOL	2020 – Present
Teaching Algorithmic Programming	Competitive & beginner students, organized events for 140+ students	2018 – Present
DeepLearn Spring School	Graph neural networks, image superresolution, instance space analysis	2021
Lajos Pósa Math Camps	Discovery learning approach, Nim games, graph coloring, countability	2013 – 2019

HONOURS & AWARDS

Competitive Programming	Codeforces master, ICPC, Olympiad of Metropolises bronze, Nemes Nationals 1st	2015 – 2023
Hackathons	1st at C2C CTF, HackZürich finalist, Special Award at HackBudapest for a SkyScanner extension	2017 – 2021
Maths	Zrínyi Ilona Nationals 1st, Bolyai Team Nationals 1st, Arany Dániel Nationals 3rd	2013 – 2019
Other	International Linguistics Olympiad bronze, Best theoretical talk at Churchill College computer science talk series	2018, 2021